

Data Analysis Memo

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What explains variation in majority formation in multiparty legislatures? There is consensus in political science about the role of district magnitude and party fragmentation (Cox 1997; Lijphart 1999; Norris 2004). However, does this rationale extend to majority formation? This question is important because it speaks to the actual performance of legislatures. It is also important because majority formation is how political and policy change occurs in democracies. If majorities are not formed in legislatures, then democracy may stagnate and fail to perform in a manner that is satisfactory for citizens. District magnitude is relatively stable, as electoral rules are a matter of contention. Comparing across cases is difficult because specific countries may have different drivers of majority formation or gridlock.

The causal logic is important. Electoral systems do not in and of themselves produce more parties, but, where multiple parties exist, they may repress the number of parties or fail to do so. District magnitude here is a key variable. Cox (1997) and Sartori before him have found that smaller district magnitudes tend to exert pressure on the number of parties, while larger magnitudes relax this pressure and allow for more proliferation.

The number of parties and their relative sizes (fragmentation, effective number of parties) may be associated to majority formation (Mainwaring 1993; Tsebelis 1995). However, fragmentation does not say anything about whether those parties have substantial reasons to agree on policy. Polarization, therefore, could also influence majority formation. While I'm not interested in the effect of polarization, it is important to consider it in a causal model.

I have chosen to focus my analysis on a single case. The first reason is that this allows me to control for noise associated to national context variation. In other words, there may be a reason why politicians in country X tend to form majorities less frequently than politicians in neighboring country Y. It could be ethnic cleavages, political culture, legislative procedural rules, or other unknown factors. Focusing on a single country allows me to isolate my variables of interest, at least to a larger measure than if I were employing a cross-country analysis.

While we know that restrictive electoral systems -such as those with smaller district magnitude- tend to repress or suppress the number of parties, it is rare that a country reduces its district magnitude. Legislators are typically in charge of electoral reform, and they reasonably understand that reducing district magnitude may act against their probability of reelection (Andrews and Jackman 2005; Negretto and Visconti 2018). Therefore, overwhelmingly, district magnitude tends to increase rather than decrease. This presents a problem for identification, as the effects of electoral system reform may linger into the future. Societal fragmentation or political polarization -or any other idiosyncratic variable that affects majority formation- may increase or decrease organically without changes in the electoral system. A better test for the effects of electoral system change on majority formation is when changes in opposite directions occur close in time. Guatemala is one such case. Since the first democratic election in the current regime in 1986, over the course of ten elections and an equal number of legislatures, district magnitude has changed 5 times, increasing 4 times and decreasing once (Brolo 2012; Brolo 2013). This within-case variation allows me to approximate a causal effect much more than in a country where magnitude has only increased. Furthermore, restricting the analysis to a single case with such a short democratic spell sets a high bar for causality: since the number of observations is very small, each single observation should be understood to have enough leverage so that changes in the dependent variable attributable to factors other than my explanation should significantly weaken my theory.

2. Causal Model and Hypotheses

My theory is as follows. District magnitude is the single most important variable that influences majority formation. First, it will increase the number of parties -presumably with different views- present in a legislature. Second, in increasing the number of parties and creating a more proportional seat allocation, it will make the legislature a more faithful reflection of polarization in society, making majority formation more difficult the more polarized society is. That is, the effect of district magnitude is sensitive to polarization: when polarization increases, the effect should be stronger. Finally, I expect majority formation to be sensitive to political events. One specific type of political event is elections. In the year following an election, when the legislature is newly integrated, they may be motivated to deliver on campaign promises and party cohesion is expected to be strongest. As the next election year approaches, legislators will be more occupied with campaigning, therefore affecting legislative output. This means that my causal model should control for distance to the prior or next election.

Hypothesis 1

Higher district magnitude is associated with lower frequency of majority formation.

Hypothesis 2

The effect of district magnitude is partially mediated by party fragmentation.

Hypothesis 3

District magnitude interacts with societal polarization to decrease the frequency of majority formation.

Hypothesis 4

Majority formation will be less frequent as the next election approaches.

I have modeled these hypothesis as follows:

Majority formation = district magnitude + party fragmentation + polarization + distance to next election + (polarization * district magnitude)

3. Data

The effect of district magnitude on majority formation would be best tested in a controlled environment by setting up n polities and assigning them different district magnitudes, observing them over a constant period of time and counting the frequency with which voting majorities form in the legislature. This is not possible. Instead, I have chosen a single polity -Guatemala-, which has some variation in district magnitude over time. This is the main independent variable. One way to measure the dependent variable -majority formation- would be to analyze voting records to identify the frequency with which each legislature achieves a majority. I don't have access to voting records, but I was able to scrape the website of the Guatemalan legislature for every piece of legislation approved since 1928. During my period of interest, each piece of legislation is officially coded with a decree number ("número de decreto") which includes a index number ranging from 1 to the number of laws passed in a year, followed by a hyphen and the year in which the legislation was passed. This allows me to count the number of pieces of legislation approved in each year. I am then using legislative output as a proxy for majority formation.

Guatemala experienced regime change several times over the course of the 20th century. However, after the democratic election in 1985, the regime has been reasonably stable, holding legislative elections which are reasonably free and fair every five years until 1995, and every four years thereafter. This allows me to use seat allocation data from each of those elections to build a party fragmentation measure. I have obtained this data from the V-Party Dataset by the V-Dem project. I have also obtained data on political polarization from the V-Dem database.

```
ggplot(data = decrees_per_magnitude, aes(x = year, y = n)) +  
  geom_col() +  
  facet_wrap(~ELECTION, scales = "free_x")
```

4. Analysis

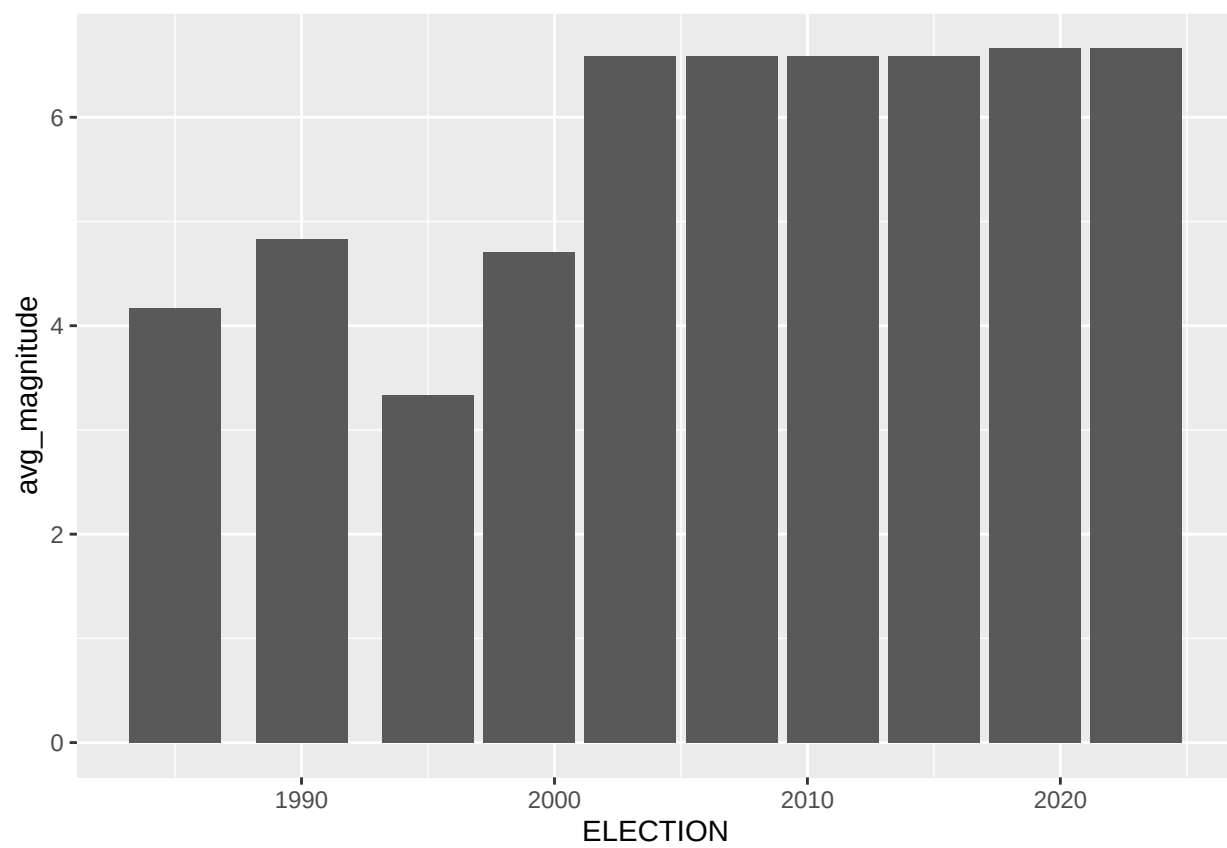


Figure 1: Average District Magnitude by Election

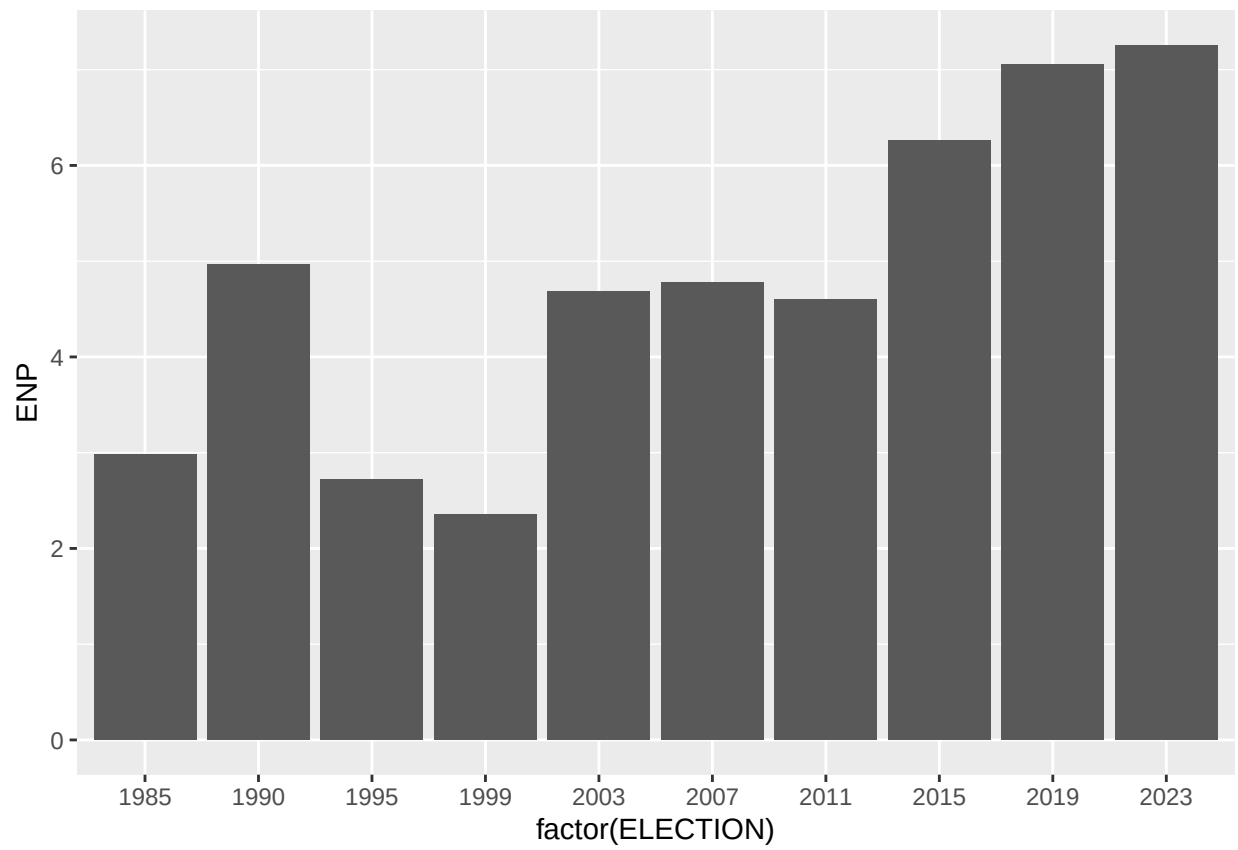


Figure 2: Effective Number of Parties by Election

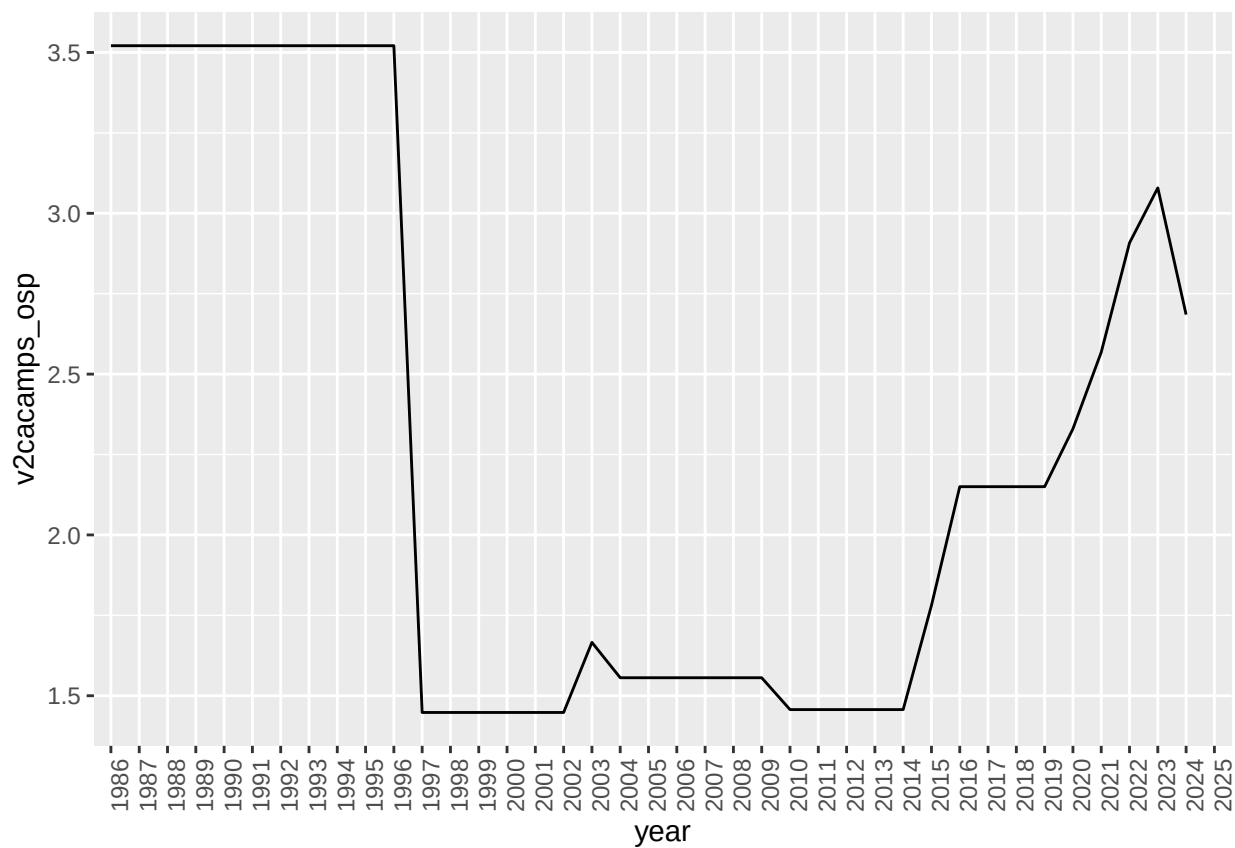


Figure 3: Polarization (v2cacamps_ops) by Year (V-Dem)

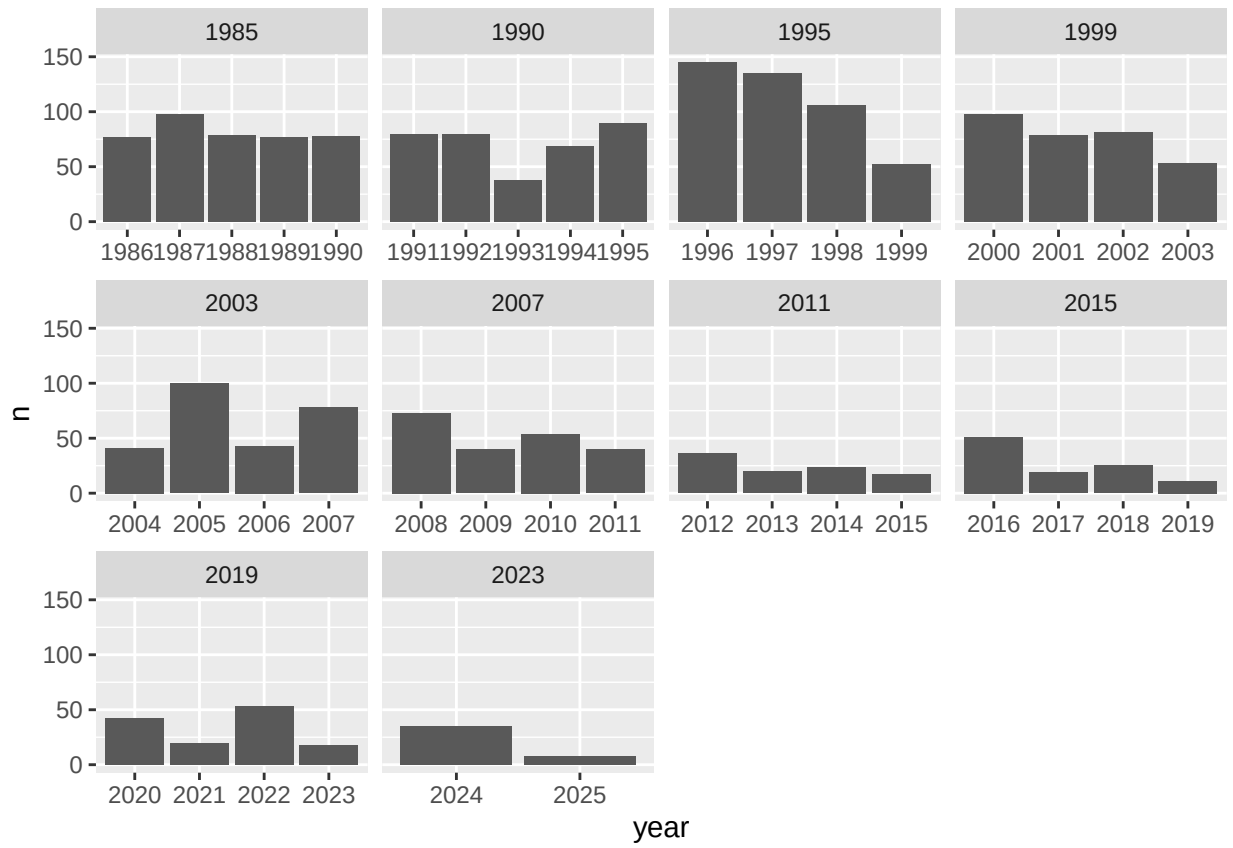


Figure 4: Number of Decrees (n) by Year

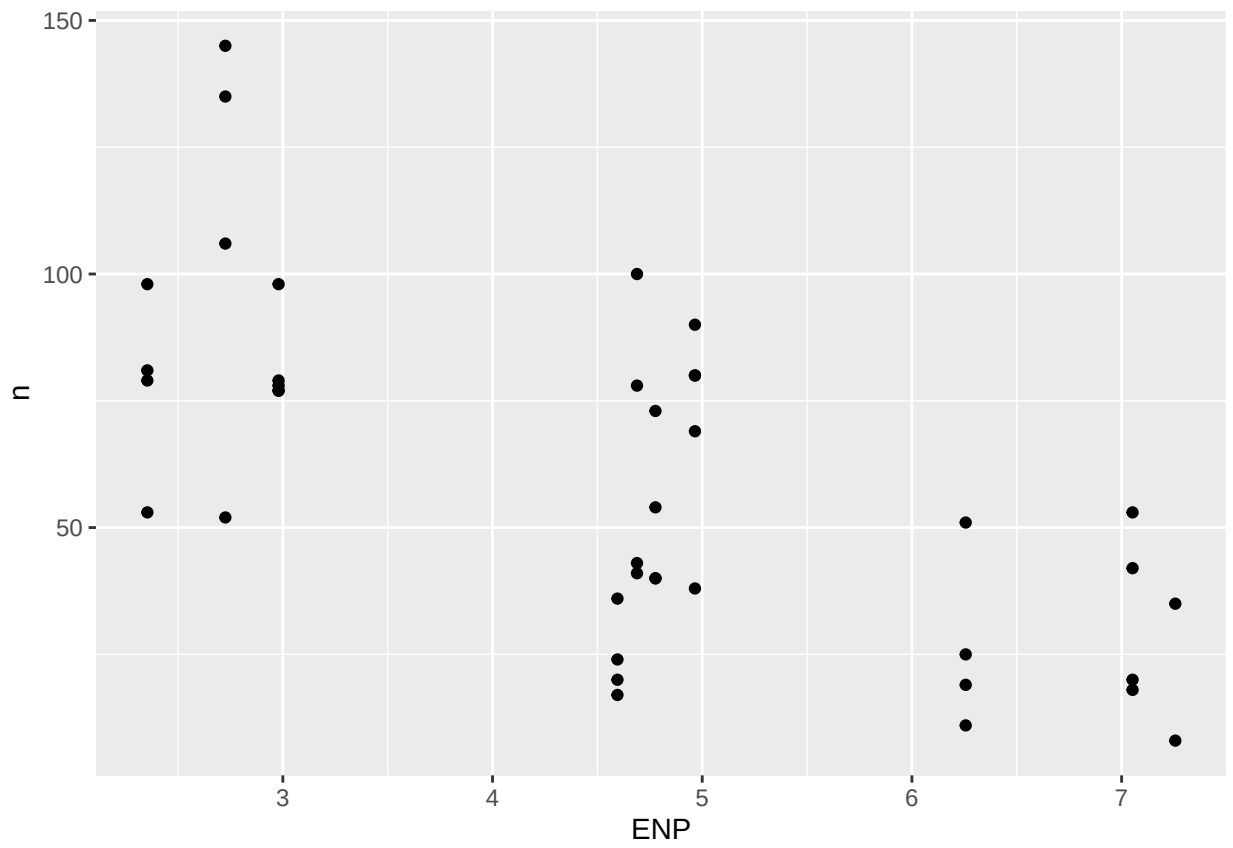


Figure 5: Number of Decrees by Effective Number of Parties

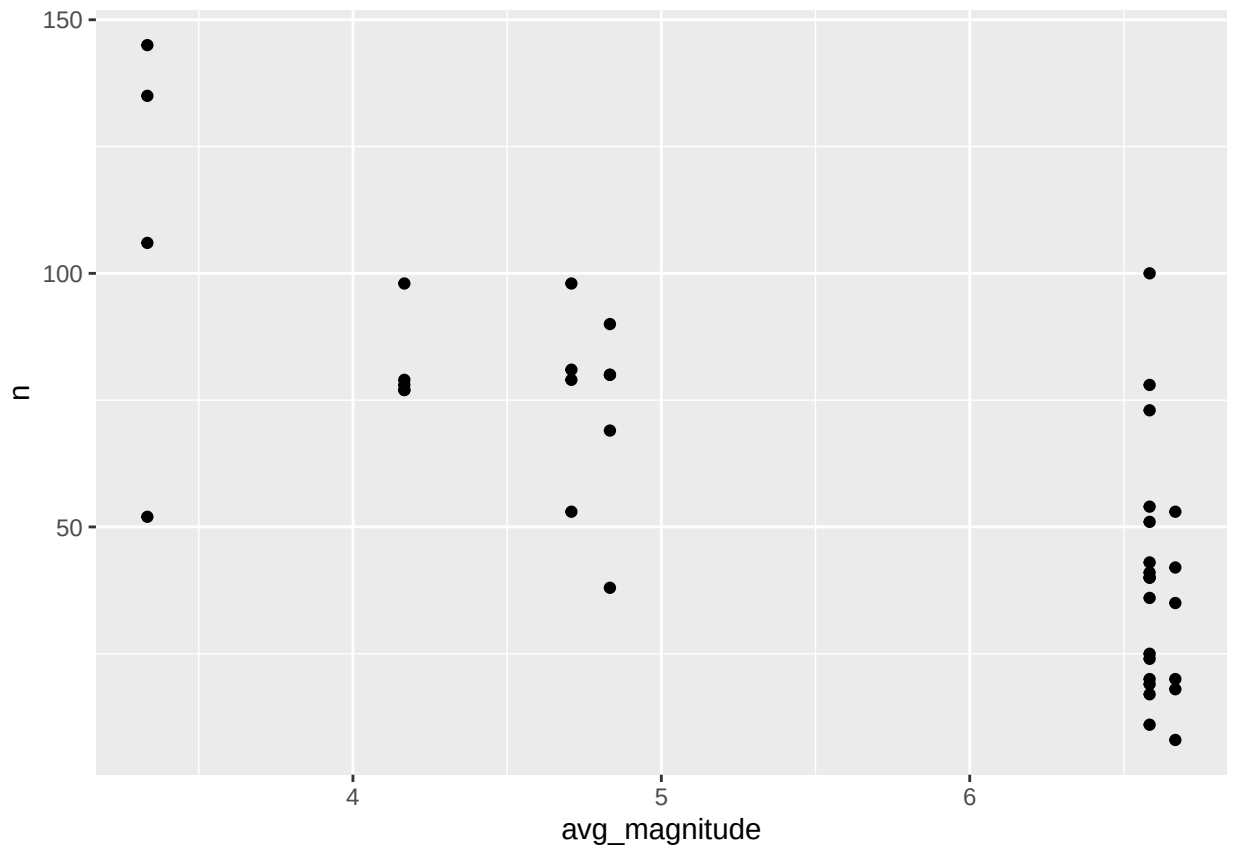


Figure 6: Number of Decrees (n) by Average District Magnitude

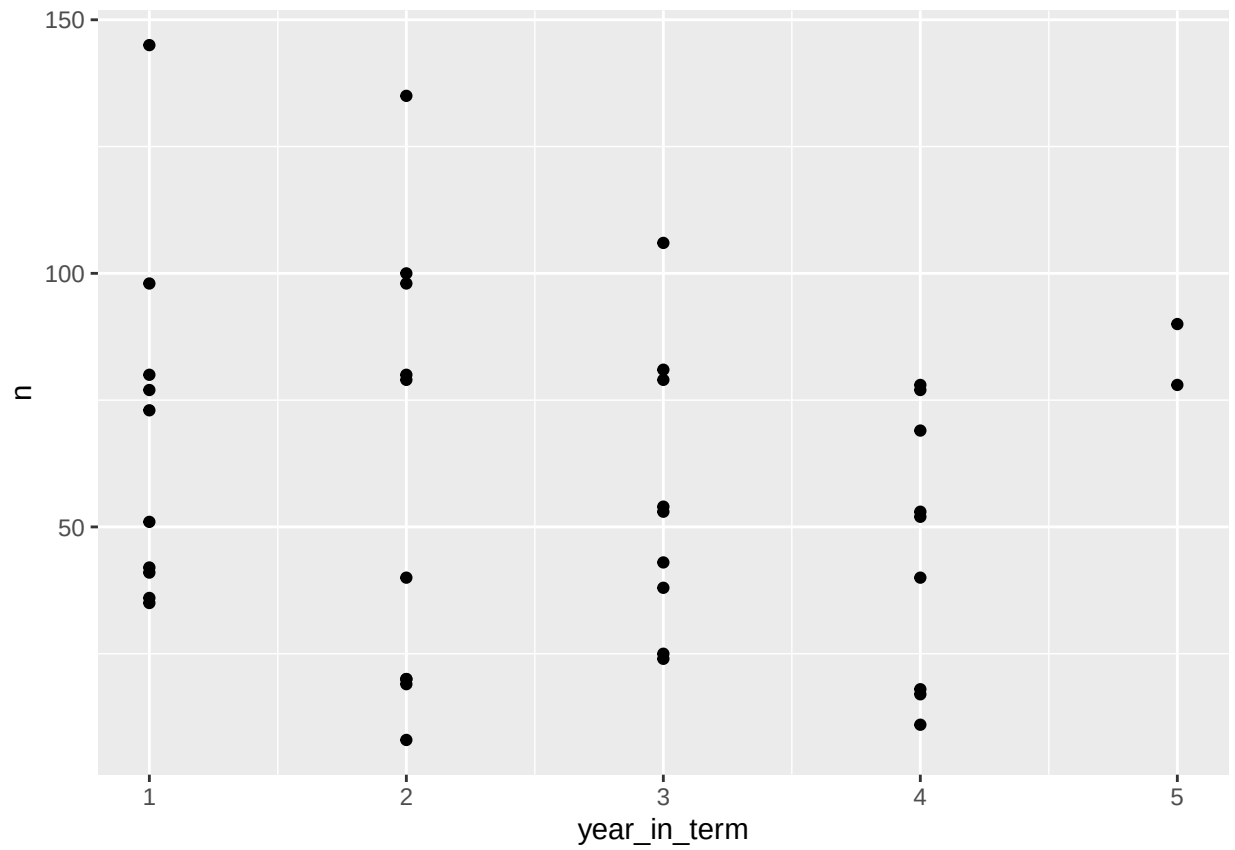


Figure 7: Number of Decrees (n) by Year in Term

I ran an OLS regression for my main independent variable -average district magnitude- and my main dependent variable -legislative output- (Model 1), and then I control for year in term and polarization in subsequent models (2 and 3). Based on my model model described above, I included an interaction term where political polarization (v2cacamps_obs) interacts with district magnitude (Model 4). I find that year in term explains a small part of legislative output across all models. In models 1-3, district magnitude is the main driver behind legislative output, accounting for more than 20% of its variation, in the predicted direction. However, in the last model, the interaction term moves all this explanatory weight to polarization, and significance disappears. Model 2 offers the most parsimonious explanation and is therefore the one I chose to continue working with below. Thus, I find support for Hypothesis 1 and Hypothesis 4, but not for Hypothesis 3.

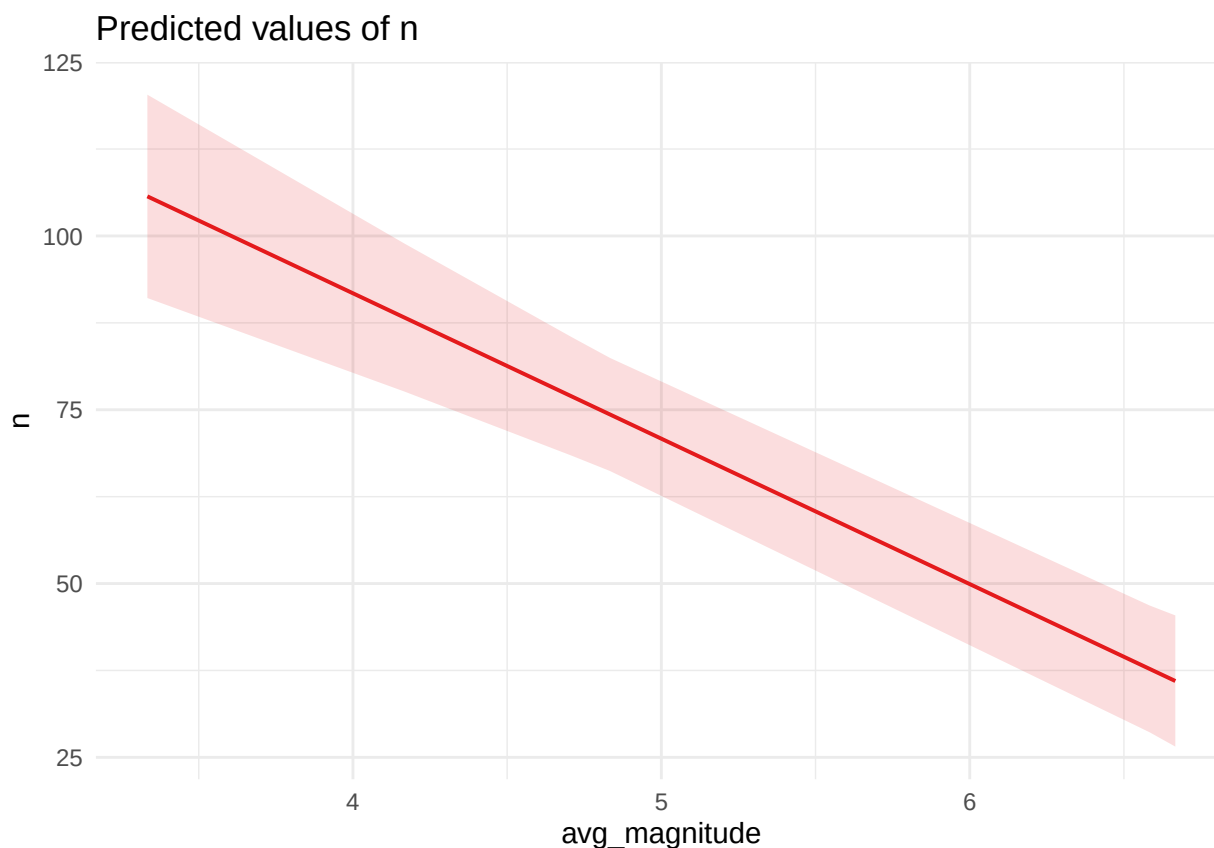


Figure 8: Legislative Output Regressed on Average District Magnitude

I then proceed to integrate the Effective Number of Parties (ENP) into the model. As we can see from the results below, the inclusion of ENP significantly reduces the weight of district magnitude. District magnitude is highly correlated with ENP, and we know from conventional literature that district magnitude impacts party fragmentation. Therefore, ENP is assumed here to be a mediating variable and is thus discarded from the model. Thus, I find support for hypothesis 2.

5. Assumption Review and Robustness Checks

The following regression diagnostics plots allow me to review the necessary assumptions for my OLS identification strategy. The Residuals vs. Fitted plot shows a reasonably stable line and identifies certain outliers (22, 20, 14). The line is slightly curved (U-Shaped) suggesting non-linearity. Similarly, in the Q-Q Residuals plot, while all points lie along the diagonal, the same points as in the Residuals vs. Fitted test lie outside of the diagonal, suggesting that residuals are may not be normally distributed. The Scale-Location plot shows reasonable uniformity in the distribution of all points, suggesting homoscedasticity. Finally, in the Residuals vs. Leverage plot, while the aforementioned outliers have more leverage than the rest of the points, they do not cross into the significance corners.

Table 1: Results of OLS Regression for Specified Model Controlling for Effective Number of Parties

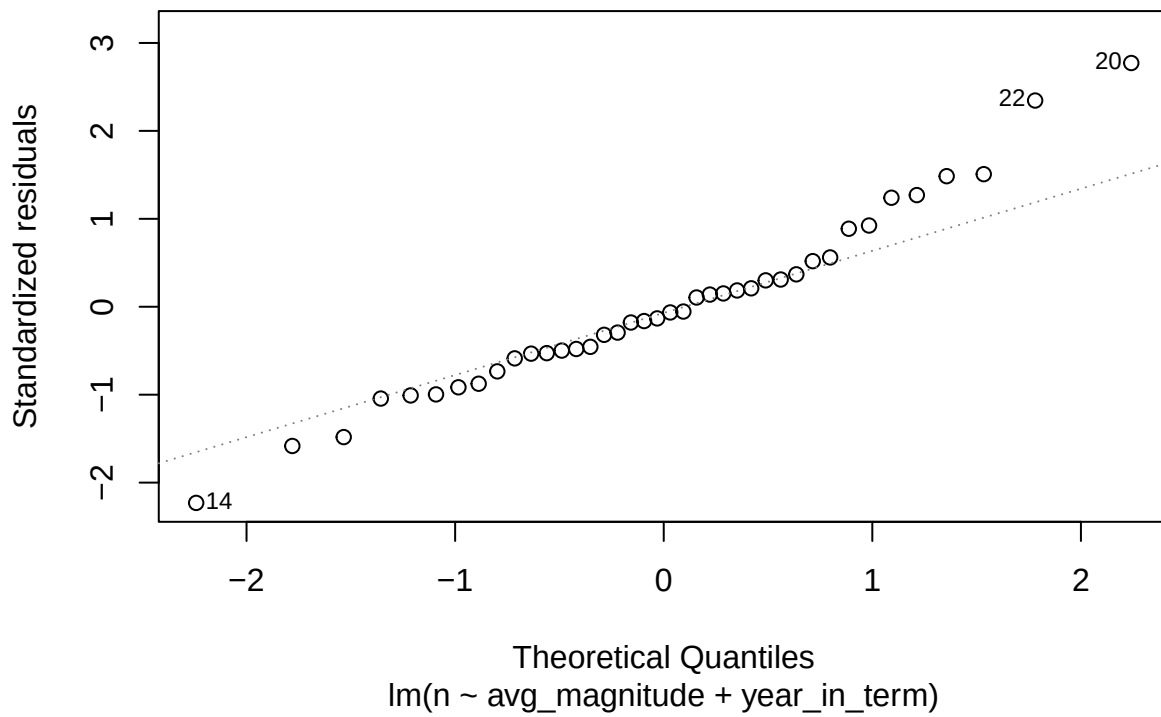
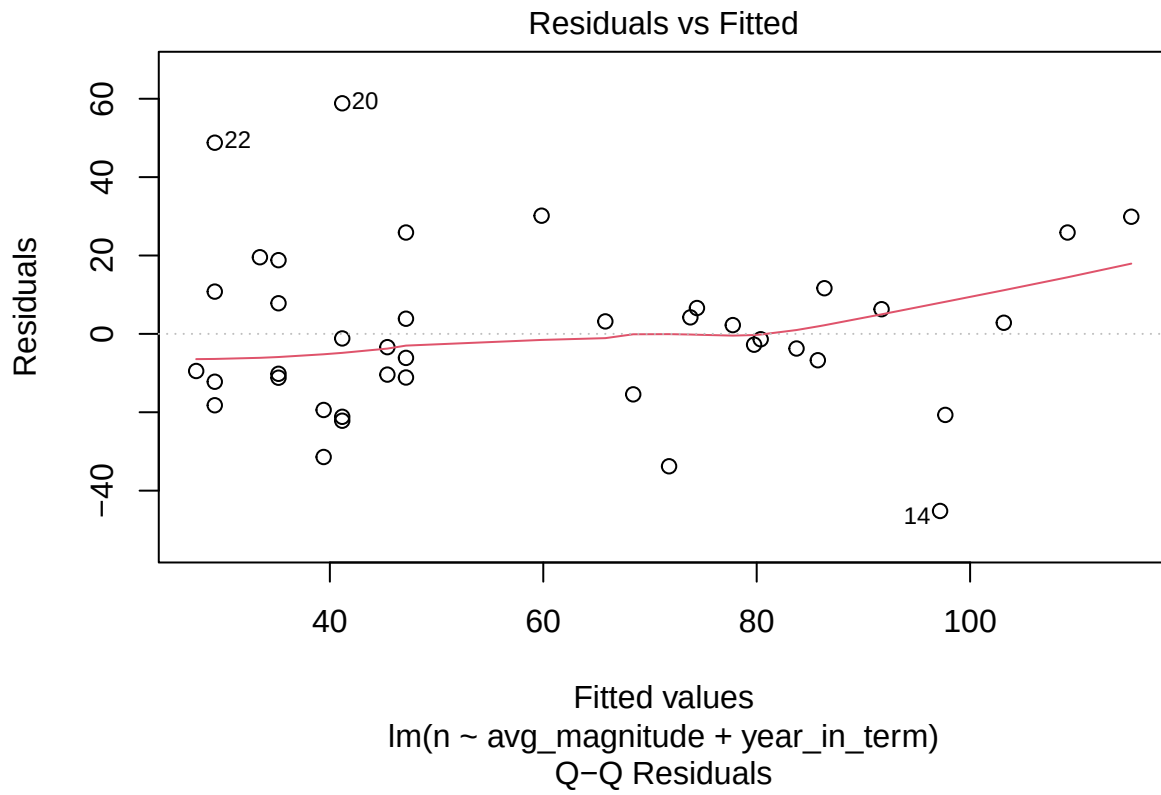
	Model 1	Model 2	Model 3
(Intercept)	166.571*** (16.944)	186.307*** (18.535)	177.600*** (29.303)
avg_magnitude	-15.235** (4.624)	-15.854*** (4.417)	-14.844* (6.269)
ENP	-4.945 (3.605)	-5.097 (3.438)	-5.154 (4.720)
year_in_term		-6.057* (2.789)	-6.328* (2.826)
v2cacamps_osp			2.037 (5.613)
Num.Obs.	40	40	39
R2	0.573	0.622	0.614
R2 Adj.	0.550	0.591	0.569
AIC	367.3	364.3	356.8
BIC	374.0	372.8	366.8
Log.Lik.	-179.626	-177.165	-172.403
F	24.821	19.780	13.545
RMSE	21.58	20.29	20.12

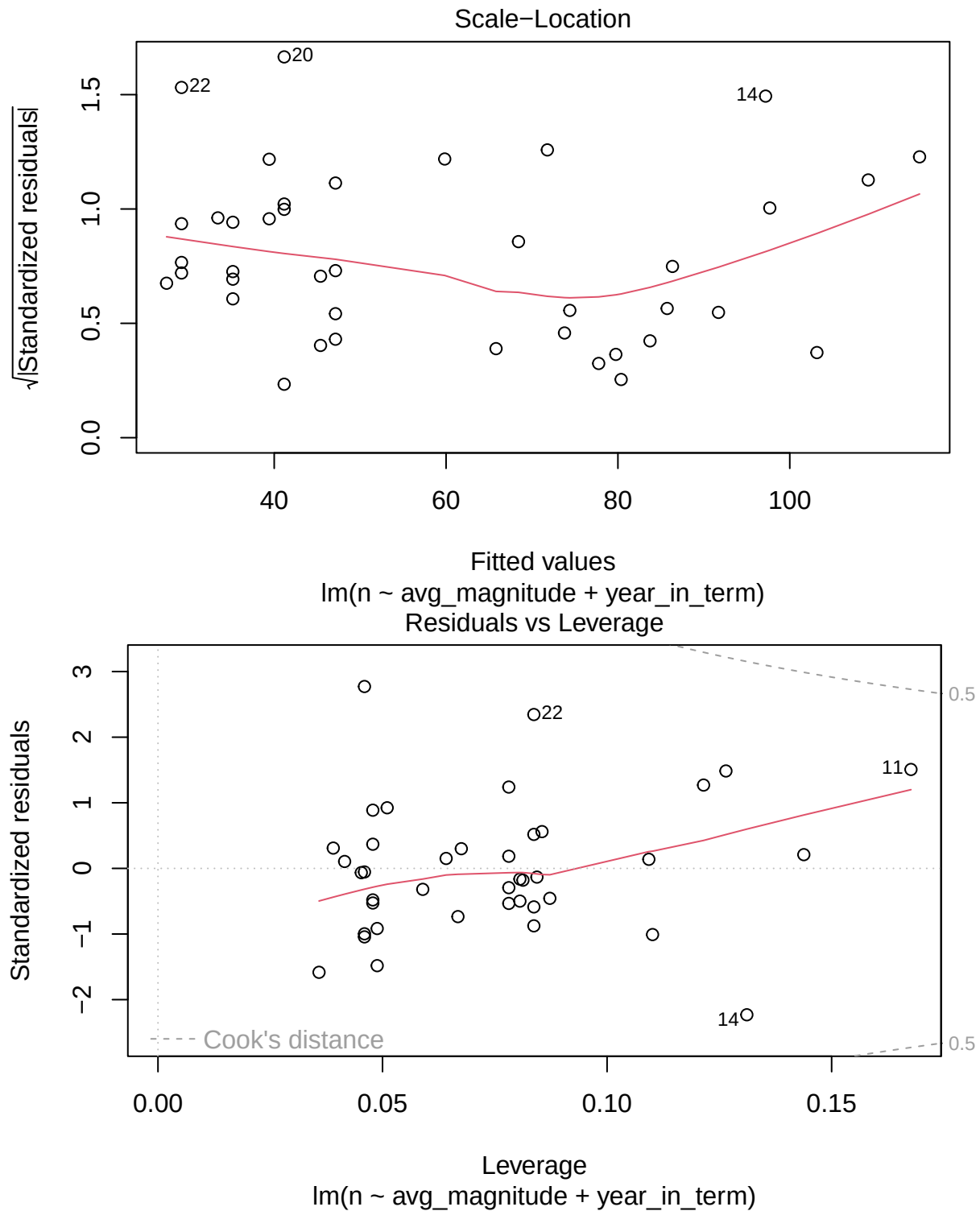
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 2: Effective Number of Parties Regressed on District Magnitude

	(1)
(Intercept)	-0.934 (0.747)
avg_magnitude	0.995*** (0.131)
Num.Obs.	40
R2	0.602
R2 Adj.	0.592
AIC	118.2
BIC	123.3
Log.Lik.	-56.110
F	57.504
RMSE	0.98

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001





Since the plots above suggest possible non-linearity and non-normality, I ran the following tests to check for each of the OLS assumptions. The results are described in Table 4. Overall, Model 2 passes all tests except for the Durbin-Watson test. Observations are not independent, mainly because one important driver of change is the integration of a new legislature with every election. Different legislatures will have different ideological configurations; in some legislatures, the president's party has a majority while in others it does not; also, different issues may come up at different times over 40 years and affect polarization, policy choices and the disposition to form majorities.

Table 4 shows the results for the same models that I specified above, adjusting standard errors to account for clustering around the variable “ELECTION”, which defines each legislative session. Coefficients and significance levels remain similar.

Table 7: Assumption Review

Assumption	Test	Statistic	P-Value	Result
Linearity	Rainbow	F = 0.563	0.891	Pass
Independence	Durbin-Watson	DW = 1.526	0.048	Fail
Homoscedasticity	Breusch-Pagan	BP = 0.387	0.824	Pass
Normality	Shapiro-Wilk	W = 0.970	0.367	Pass
Multicollinearity	VIF	avg_magnitude = 1.016, year_in_term = 1.016	—	Pass

7. Bibliography

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Data Sources

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